

Class #1 - Eigenvectors and Eigenvalues

1. Example: Consider the system of DE

$$\frac{dx(t)}{dt} = Ax(t).$$

Solution to above depends on the eigenvalues of the matrix A .

2. Def: Given a matrix $A \in$ _____, a _____ vector x such that $Ax = \lambda x$ for some _____ λ is called an _____ of _____.
3. Example: Let A be an $n \times n$ zero matrix. Find an eigenvector. Is $\lambda = 1$ an eigenvalue of A ?

4. A scalar λ is called an _____ of A if there is a _____ solution to the equation

$$Ax = \lambda x.$$

That nontrivial solution is called an _____ of A corresponding to _____.

5. Example: Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$? If so, find the eigenvalue.

6. Example: Is $\lambda = 4$ an eigenvalue of $\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$? If so, find one corresponding eigenvector.

7. Note: λ is an eigenvalue of if the equation _____, i.e., _____ has a _____ solution.
Hence, λ is an eigenvalue of A if and only if columns of the matrix _____ are _____, i.e., _____ = 0.

8. Def: The set of all solutions to the equation

$$(A - \lambda I)x = \underline{\hspace{2cm}}$$

is called *eigenspace of A corresponding to λ* .

9. Note: eigenspace of A corresponding to $\lambda = \underline{\hspace{2cm}}$ $(A - \lambda I)$ which is a subspace of _____.

10. Note: $\dim$ eigenspace of A corresponding to $\lambda = \text{nullity of } \underline{\hspace{2cm}}$, and this number is called the _____ *multiplicity of λ* .

11. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$. Without any calculation, show that $\lambda = 0$ is an eigenvalue of A . Find a basis for the eigenspace corresponding to $\lambda = 0$. What is the geometric multiplicity of $\lambda = 0$?
Fill in the blank: Any _____multiple of an eigenvector is also an _____corresponding to the same _____.

12. Eigenvalues of a triangular matrix:

Fill in the blank: If A is an $n \times n$ triangular matrix, then a _____element is an eigenvalue of A .